# IRU 600

## ALL-INDOOR MICROWAVE RADIO

DATASHEET [ANSI]

When you can't sacrifice reliability for high capacities, the Aviat IRU 600 represents the ultimate high performance, reliability, flexibility, and the highest capacity solution for Mission Critical applications. The IRU 600 is the industry's most secure all-indoor radio (FIPS 140-2), with market-leading system gain to enable smaller antennas and lowest TCO.

### Highest Power Digital Radio Ever

IRU 600 incorporates the very latest in RF technology to enable break-through RF performance with highest reliability. It's Extra-High Power (EHP) and Ultra High Power (UHP)[4] options support more transmitter power than any alternative solution, extending path distances, unlocking more capacity, and enabling smaller, less expensive antennas to be used to reduce tower related costs.

### Highest Capacity

The IRU 600 incorporates the latest features to enable multi-gigabit capacity links, including support for wide channels, 4096QAM modulation, XPIC, ACCP, ACAP and advanced Layer 1 Link Aggregation for multi-channel bonding, and now 6+11 GHz Multi-Band options, which all result in link capacities that are 3x higher or more than previous product technologies.

### Tough, Durable and Secure

The IRU 600 is the most secure microwave radio on the market today and with FIPS 140-2 validation, mission critical networks are always protected. Tough, durable, and dependable, and now, thanks to a new upgrade with the latest in RF technology, IRU 600 is prepared for the data demands of mission critical networks of the future.

PROUDLY MANUFACTURED IN THE USA

Comprehensive platform coverage, including support for FCC, NTIA and ISED Canada frequency bands, and for all-IP, all-TDM (native or PWE), or hybrid TDM+IP applications.

Ultra-high transmit output power up to +38.5 dBm for 6GHz and +38 dBm with 11 GHz (guaranteed at the antenna port) or 5 times more than high power than previous generation.

Flexible and configurable radio configuration options, including expansion ports to suit every network need and ensure easy future scalability/expansion.

## System Parameters

| System | Frequency Range (GHz) | TR-Spacings Supported (MHz) Non-standard T-R options supported with spot tuned filters | Standard Frequency Assignments Supported per Regulatory Plans |
|---|---|---|---|
| 4 GHz | 4.400-5.000 | 160 (EHP only) 300, 312 | NTIA Red Book<br>ITU-R F.1099-5 |
| 5.8 GHz | 5.727.5 - 5.847.5 | ≥65 | FCC Part 15.407<br>RSS-247 Issue 2 |
| L6 / U6 GHz | 5.925 - 6.425,<br>6.525 - 6.875,<br>6.425 - 6.930 | 150, 160,<br>170, 180,<br>240, 252.04,<br>266, 340, 345 | FCC Part 101<br>SRSP 305.9<br>SRSP 306.4 |
| FCC 7 | 6.875 - 7.125 | 150 | FCC Part 74, 101 |
| 7 GHz | 7.125 - 7.750 | 150, 175, 300, Unpaired | NTIA Red Book<br>SRSP307.1 |
| 8 GHz | 7.750 - 8.500 | 300, 360, Unpaired | NTIA Red Book<br>SRSP 307.7<br>SRSP 308.2 |
| 10.5 GHz | 10.500 - 10.680 | 65 | FCC Part 101<br>SRSP 310.5 |
| 11 GHz | 10.700 - 11.700 | 490, 500 | FCC Part 101<br>SRSP 310.7 |

| General | | |
|---|---|---|
| Frequency Band options | | 4, 5.8, L6, U6, FCC 7, 7, 8, 10.5 and 11 GHz |
| Modulation and Coding Options | Fixed and Adaptive | QPSK, 16, 32, 64, 128, 256, 512, 1024, 2048 and 4096 QAM |
| Channel Sizes Supported | | 3.75, 5, 10, 12.5, 20, 25, 30, 40, 60 and 80 MHz |
| Capacity Support | Airlink Capacity<br>Ethernet / IP Throughput<br>Native TDM / OC3 | 8-716 Mbit/s<br>15-940 Mbit/s<br>5 to 127 DS1, 2xOC3, 6xDS3 |

## Electrical and Mechanical

| Size / Weight (1+1 configuration) | ACU+Chassis+2RFU (includes waveguide terminator) | Chassis+2 RFU |
|---|---|---|
| 4G SP/HP units | 3RU: 5.2 in x 19 in x 13.18 in<br>(132 mm x 482 mm x 335 mm)<br>33 lb (15 kg) | 3RU: 5.2 in x 19 in x 13.18 in<br>(132 mm x 482 mm x 304 mm)<br>27.5 lb (12.5 kg) |
| 4G EHP units | 3RU: 5.2 in x 19 in x 13.18 in<br>(132 mm x 482 mm x 335 mm)<br>33 lb (20 kg) | 3RU: 5.2 in x 19 in x 13.18 in<br>(132 mm x 482 mm x 304 mm)<br>37.5 lb (17 kg) |
| 5.8 – 11 GHz SP/HP units | 2RU 3.5 in x 19 in x 16.25 in<br>(88 mm x 482 mm x 412 mm)<br>32.5 lb (14.7 kg) | 2RU 3.5 in x 19 in x 12 in<br>(88 mm x 482 mm x 304 mm)<br>28.0 lb (12.7 kg) |
| L6 – 11 GHz EHP units | 2RU 3.5 in x 19 in x 16.25 in<br>(88 mm x 482 mm x 412 mm)<br>33.9 lb (15.4 kg) | 2RU 3.5 in x 19 in x 12 in<br>(88 mm x 482 mm x 304 mm)<br>29.5 lb (13.4 kg) |

| Antenna Interface | 4 GHz | 5.8/L6/U6/FCC 7 GHz | 7/8 GHz | 10.5/11 GHz |
|---|---|---|---|---|
| ACU Main / Expansion Antenna Port Interface | CMR-187 | CMR-137 | CMR-112 | UG 39 |
| Extension Kit Antenna Port Interface (near top of rack) | CPR-187G | CPR-137G | CPR-112G | CPR-90G |

| Environmental | | |
|---|---|---|
| Operating Temperature | Guaranteed | -10° to +55° C (14° to +131° F) |
| Humidity | | 0 to 95%, non-condensing |
| Altitude | | 15,000 ft (4,500 m) AMSL |

| Standards Compliance | |
|---|---|
| EMC | FCC CFR 47, Part15, ICES-003 |
| Operation, Storage, Transportation | EN 300 019, Class 3.2<br>EN 300 019, Class 1.2<br>EN 300 019, Class 2.3 |
| Safety | UL 62368-1<br>CAN/CSA C22.2 No. 62368-1-14 |
| Electric Power Substations [1] | IEEE 1613 |
| Radio Frequency | FCC Part 15.407, ISED RSS-247, Issue 2 (5.8 GHz only),<br>FCC CFR47, Part 101, ISED Canada's SRSPs (L6 to 11GHz) |
| NEBS [2] | GR-1089-CORE,GR-63-CORE |
| Security (with INUe) | FIPS 197 validated (Certificate #C5)<br>FIPS 140-2 validated (Certificate #4187) |
| DoDIN APL Certification (with INUe) | APLITS Tracking Number: 2032201 |

| IF Specifications | | |
|---|---|---|
| IF Frequency | Transmit/Receive | 311 MHz/126MHz |
| IF Cable Length | | 1000 ft (300 m) max (limitation apply) |
| Test Point | RSSI Monitoring Point<br>TX Monitoring Port | Test Points female<br>SMA female |

| Emission Designators | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| MHz | 3.75 | 5 | 10 | 12.5 | 20 | 25 | 30 | 40 | 60 | 80 |
| QPSK | 3M75G7W | 5M00G7W | 10M0G7W | 12M5G7W | 20M0G7W | 25M0G7W | 30M0G7W | 40M0G7W | 60M0G7W | 80M0G7W |
| QAM | 3M75D7W | 5M00D7W | 10M0D7W | 12M5D7W | 20M0D7W | 25M0D7W | 30M0D7W | 40M0D7W | 60M0D7W | 80M0D7W |

| Transmitter Specifications | | | | |
|---|---|---|---|---|
| Transmit Power Tolerance | -0 +1 dB | | | |
| Transmitter Source | Synthesized | | | |
| Frequency Stability | ±5 ppm | | | |
| Manual Transmitter Power Control Range [3] | Configurable in 0.1 dB steps from min to max power levels<br>(Refer to TX Power Specifications)<br>Resolution: 0.1 dB | | | |
| | 4G | 4G EHP | 5.8G – 11G MP/EHP | 11G UHP |
| QPSK | 17.0 dB | 15dB | 20.0 dB | 18 dB |
| Automatic Transmitter Power Control | Range<br>Resolution /<br>Speed | Configurable over full available manual attenuation range<br>0.1 dB steps / 6 dB per second | | |
| Synthesizer Resolution | 5 KHz | | | |
| Channel Selection | By softwarecontrol withintuning range of RFU | | | |
| Transmitter Mute | > 50 dB | | | |

| Receiver Specifications | | |
|---|---|---|
| Receiver Source | | Synthesized |
| Frequency Stability | | ±5 ppm |
| Receiver Overload | BER = 1E-6<br>No damage | -22 dBm<br>0 dBm |
| Residual (Background) Bit Error Rate | | Better than 1E-13 |
| RSSI Accuracy [3] | -40 to -70 dBm, +0 to +35°C<br>-25 to -85 dBm, -10 to +55°C | ± 2 dB<br>± 4 dB |

| Interfaces | | |
|---|---|---|
| DC power connector | | 2 pin D-SUB Power Type |
| IF cable connector | | SMA |
| Test point | RSSI Monitoring Point<br>TX Monitoring Port | Test Points female<br>SMA female |
| Antenna port Interface | | Standard EIA rectangular waveguide |
| Polarization | | Vertical (standard) or Horizontal |

EHP L6 and U6 unpaired frequencies usage: see the Eclipse User Manual – for "Small non-standard T/R spacings".

[1] Complies with the exception of clause 3.1.1 (operational temperature range) and clause 10 (device cooling).

[2] IRU 600v4 complies in all areas except for R9-16 and R10-5, GR-1089-CORE. INU complies in all areas except for R10-5, GR-1089-CORE and R4-34, O4-35, O4-36, GR-63-CORE.

[3] RSSI accuracy applies when there is no potential interferer signal present within +/- 10MHz of the Rx. Frequency.

## Disclaimer

This material is for informational purposes only and does not constitute a legal obligation to deliver any product, feature or functionality and should not be relied upon in making purchasing decisions. All specifications are guaranteed values, at room temperature (20 to 30ºC, 68 to 86ºF), referenced to the ACU antenna port (including ACU losses) unless otherwise stated, and are subject to change without notice. The development, release and timing of any features or functionality described for our products is at Aviat Networks' sole discretion.

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